

## Lecture 6

### Diagonalization, Cayley–Hamilton, Functions of Operators

---

Throughout,  $T \in \mathcal{L}(V)$  and  $\chi_T(\lambda) = \det(\lambda I - T)$  is the characteristic polynomial. Tags mark the flavour; a ★ marks a harder one.

#### Diagonalization

**Exercise 6.1** (computational) Diagonalize  $A = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$ : find  $P$  and  $D$  with  $P^{-1}AP = D$ .

**Exercise 6.2** (computational) Decide whether  $A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$  is diagonalizable, with reasons.

**Exercise 6.3** (computational) For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  (eigenvalues 3, 1, eigenvectors  $(1, 1)$ ,  $(1, -1)$ ) compute  $A^n$  for general  $n$ .

**Exercise 6.4** (computational) Diagonalize  $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  over  $\mathbb{R}$ : find  $P$  and  $D$  with  $P^{-1}AP = D$ .

**Exercise 6.5** (computational) Diagonalize  $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$  over  $\mathbb{C}$ .

**Exercise 6.6** (computational) Decide whether  $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$  is diagonalizable, with reasons.

**Exercise 6.7** (computational) For  $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  compute  $A^n$  for general  $n \geq 0$ .

**Exercise 6.8** (proof) Suppose  $T \in \mathcal{L}(V)$  is diagonalizable and has only one eigenvalue  $\lambda$ . Prove that  $T = \lambda I$ .

#### Cayley–Hamilton

**Exercise 6.9** (computational) Use the Cayley–Hamilton theorem to express  $A^{-1}$  as a polynomial in  $A$  for  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ .

**Exercise 6.10** (computational) Use the Cayley–Hamilton theorem to express  $A^{-1}$  as a polynomial in  $A$  for  $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ .

**Exercise 6.11** (computational) For  $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$ , use Cayley–Hamilton to write  $A^4$  in the form  $\alpha A + \beta I$ , and give the resulting matrix.

**Exercise 6.12** (computational) For  $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$  compute  $\chi_A$  and use Cayley–Hamilton to express  $A^{-1}$  as a polynomial in  $A$ ; give  $A^{-1}$  explicitly.

**Exercise 6.13** (proof) Suppose  $T \in \mathcal{L}(V)$  is invertible on a complex space of dimension  $n$ . Prove that  $T^{-1}$  is a polynomial in  $T$ .

**Exercise 6.14** (proof) Let  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . By direct computation show that  $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$ , thereby verifying Cayley–Hamilton for every  $2 \times 2$  matrix.

## Functions of operators

**Exercise 6.15** (computational) For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$  compute  $e^{tA}$ .

**Exercise 6.16** (proof) Suppose  $T$  is diagonalizable with  $T = PDP^{-1}$ . Prove that for any polynomials  $p, q$ , the operators  $p(T)$  and  $q(T)$  commute.

**Exercise 6.17** (computational) Let  $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ . Compute  $e^{tA}$  and describe the solution curves of  $\dot{x} = Ax$ .

**Exercise 6.18** (★) Suppose  $T$  is diagonalizable with eigenvalues in  $\{0, 1\}$  (a projection-like operator). Prove that  $T^2 = T$ .

**Exercise 6.19** (★) Suppose  $A$  is diagonalizable and every eigenvalue of  $A$  satisfies  $|\lambda| < 1$ . Prove that  $A^n \rightarrow 0$  as  $n \rightarrow \infty$  (entrywise).

**Exercise 6.20** (computational) Write  $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  as  $A = D + N$  with  $D$  diagonalizable,  $N$  nilpotent,  $DN = ND$ , and use it to compute  $e^{tA}$ .

**Exercise 6.21** (computational) For  $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  compute  $e^{tA}$ .

**Exercise 6.22** (computational) Find the positive square root  $\sqrt{A}$  of  $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$  (the one with positive eigenvalues), so that  $(\sqrt{A})^2 = A$ .

**Exercise 6.23** (computational) Compute  $e^{tA}$  for  $A = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$ .

**Exercise 6.24** (computational) Solve  $\dot{x} = Ax$  with  $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$  and  $x(0) = (2, 0)$ .

**Exercise 6.25** (computational) Compute  $e^{tA}$  for  $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ .

**Exercise 6.26** (computational) Let  $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ , with  $\text{spec}(A) = \{1, 3\}$ . Using spectral mapping, find  $\text{spec}(B)$  for  $B = A^3 - 2A + I$ .

**Exercise 6.27** (★) On the spin-1 representation the raising operator acts by  $J_+|-1\rangle = \sqrt{2}|0\rangle$ ,  $J_+|0\rangle = \sqrt{2}|1\rangle$ ,  $J_+|1\rangle = 0$ . Write its matrix in the basis  $(|1\rangle, |0\rangle, |-1\rangle)$  and show  $J_+$  is nilpotent.

**Exercise 6.28** (proof) Let  $A$  be diagonalizable. Prove that  $\det(e^A) = e^{\text{tr} A}$ .